Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
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MYCOBACTERIUM TUBERCULOSIS GENOME SEQUENCING REPORT

NOT FOR DIAGNOSTIC USE
Page 1 of 2 Patient ID: 12345678910 | Date: 2017-01-01 | Location: Someplace
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Patient Name John Doe Barcode

Birth Date 2000-01-01 Patient ID 12345678910

Location Someplace Sample Type Sputum

Sample Source Pulmonary Sample Date 2016-12-25

Sample ID A12345678 Sequenced From MGIT Cultured Isolate

Reporting Lab Lab Name Report Date/Time 2017-01-01, 15:36

Requested By Requester Name Requester Contact requester@email.com

Summary

The specimen was positive for **Mycobacterium tuberculosis**. It is **resistant to isoniaizd and rifampin**. It belongs to a cluster, suggesting **recent transmission**.

Organism The specimen was positive for ***Mycobacterium tuberculosis***, lineage 2.2.1 (**East-Asian Beijing**).

Drug Susceptibility

=2pt lightgray to X[1.1,l]X 4*Resistance is reported when a high-confidence resistance-conferring mutation is detected. **"No mutation detected" does not exclude the possibility of resistance**. No drug resistance predicted
Mono-resistance predicted
[HTML]EFEFEF Multi-drug resistance predicted
Extensive drug resistance predicted

=5pt lightgray to X[0.5,l]X[0.75,l]X[0.75,l]X[2,l]

Drug class	Interpretation	Drug	Resistance Gene (Amino Acid Mutation)
		Ethambutol	No mutation detected
-2*Susceptible		Pyrazinimide	No mutation detected
[HTML]EFEFEF		[HTML]EFEFEFIsoniazid	[HTML]EFEFEFkatG (S315T)
-4*First Line	-2*[HTML]EFEFEFResistant	[HTML]EFEFEFRifampin	[HTML]EFEFEFrpoB (S531L)
		Streptomycin	No mutation detected
		Ciprofloxacin	No mutation detected
		Ofloxacin	No mutation detected
		Moxifloxacin	No mutation detected
		Amikacin	No mutation detected
		Kanamycin	No mutation detected
-7*Second Line	-7*Susceptible	Capreomycin	No mutation detected

Cluster Detection The current isolate was clustered with previously sequenced isolates, suggesting **recent transmission**.

=5pt lightgray to XX Relatedness Number of prior matching isolates

Closely Related (< 5 mutations apart) **2** isolates
Related (6 to 30 mutations apart) **6** isolates

Assay Details =5pt lightgray to |XU||XU|

Sample ID A12345678 Barcode

Sequencer Illumina HiSeq 2500 Method WGS

Pipeline ReSeqTB v.3.2c Reference H37Rv

Comments

No additional comments for this report

gray***Standard Disclaimer:*** *Low frequency hetero-resistance below the limit of detection by sequencing may affect typing results. The interpretation provided is based on the current understanding of genotype-phenotype relationships.*

Authorised =5pt lightgray to |XU||XU|

Signature Name

Position Date